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भारतीय मानक मसौदा
डेंटल इंप्रेशन (मॉडलिंग) कंपाउंड के लिए विशिष्टि
(IS 6038 का दूसरा पुनरीक्षण)

Draft Indian Standard

**SPECIFICATION FOR DENTAL IMPRESSION
(MODELLING) COMPOUND**
(Second Revision of IS 6038)

[ICS 11.060.20]

Dentistry Sectional Committee, MHD 08

Last date for comments:
10 Sep 2026

FOREWORD

(Formal clauses will be added later)

This Indian Standard was originally published in 1970 and the first revision was published in 1979. The second revision has been brought out to align cross references of Indian Standards and revised certification clauses.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off in accordance with IS 2: 2022 ‘Rules for rounding off numerical values (second revision)’.

1 SCOPE

This standard prescribes the requirements and the methods of sampling and test for dental impression (modelling) compounds.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No</i>	<i>Title</i>
IS 4905 :2015 ISO 24153 : 2009	Random sampling and randomization procedures (<i>First Revision</i>)

3 TYPES

3.1 The material shall be of the following three types.

3.1.1 *Type 1* – called ‘impression compound’, suitable for taking impressions in the oral cavity.

3.1.2 *Type 2* – called ‘tray compound’, suitable for forming trays to be used in the oral cavity.

3.1.3 *Type 3* – called ‘tracing stick’, or ‘low heat compound’, suitable for taking impressions in the oral cavity.

4 REQUIREMENTS

4.1 Description

The material shall be homogeneous throughout, capable of giving a smooth, glossy surface after flaming and shall also give firm and smooth margins after trimming with a knife at room temperature.

4.2 Material

The material shall essentially be a composition of resins and fillers with necessary additives, such as plasticizers and colouring agents.

4.3 Forms

4.3.1 The material of Type 1 shall be furnished in cake or stick form, and shall be uniform and suitable for taking impression in the oral cavity.

4.3.2 The material of Type 2 shall be furnished in cake form and shall be uniform and suitable for forming trays to be used in the oral cavity.

4.3.3 The material of Type 3 shall be furnished in stick form and shall be uniform and suitable

for use in corrected impression techniques.

4.4 Colour

The material of Type 1 shall be red; Type 2 black; and, Type 3 green in colour.

4.5 Odour and Flavour

The material shall have no unpleasant odour or flavour.

4.6 Irritation

The material shall not irritate oral tissues and shall not contain poisonous ingredients in such concentration as may be harmful to human beings when used as directed.

4.7 Working Temperature Range

The material shall soften to a workable consistency within the temperature range specified below when tested as prescribed in A-1:

Type 1	53.5 to 56.5°C
Type 2	55.0 to 58.0°C
Type 3	50.0 to 52.2°C

4.8 Working Time Range

The working time range of the material shall be from 50 to 80 seconds when tested as prescribed in A-1.

4.9 Flow

The percent of flow shall be as given in Table 1 when tested as prescribed in A2.

TABLE 1 FLOW TEST (Clause 4.9)			
Temperature	Flow, Percent		
°C	r	h	
	Type 1	Type 2	Type 3
(1)	(2)	(3)	(4)
37.0	6.0 <i>Max</i>	2.0 <i>Max</i>	6.0 <i>Max</i>
45.0	88.0 <i>Max</i>	70.0 to 85.0	88.0 <i>Min</i>

4.10 Impression Test

The material of Type 1 and Type 3 shall accurately record the complete surface detail of the test block (see Fig. 3) at 45°C when tested as prescribed in A-3.

5 TEST METHODS

Tests shall be conducted according to methods prescribed in Appendix A.

6 SAMPLING

Representative samples of the material shall be drawn as prescribed in Appendix B or as agreed to between the purchaser and the supplier.

7 PACKING

Each cake or each stick shall be separated with, or silicon paper and packaged in a carton of appropriate size. Each carton shall be securely closed to prevent spillage of contents under normal handling.

8 MARKING

8.1 Each carton shall bear legibly and indelibly the following information:

- a) Name and type of the material;
- b) Name of the manufacturer and his recognized trade-mark, if any;
- c) Gross and net mass;
- d) Number of cakes/sticks in each carton;
- e) Size of the cakes/stick;
- f) Batch number; and
- g) Date of manufacture.

8.2 Manufacturer's Instruction

The manufacturer shall supply each package instructions for use of the material. These instructions shall include the method of softening, working temperature and a curve or data showing the shrinkage of the compound from 40 to 20°C.

9 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

APPENDIX A
(Clause 5)
METHODS OF TEST

A-1. WORKING RANGE TEST

A-1.1 Apparatus

A-1.1.1 Water-Bath

A-1.2 Procedure - Adjust and maintain the water-bath at the desired working temperature range (see 4.7). Take 30 ± 2 g of the material, place in the water-bath and warm for 15 minutes. Remove the warmed material from the water-bath and work (knead) in wet hand for 15 seconds. Return the material to water-bath and condition for an additional 5 minutes. Remove the material from the water-bath again and work (knead) in wet hand until the material loses plasticity or proper working consistency and can no longer be moulded by finger pressure. The total time elapsed from final removal from the water-bath to the loss of plasticity shall be termed as working time.

A-2 FLOW TEST

A-2.1 Apparatus,

A-2.1.1 Flow Testing Machine- The flow testing machine (see Fig. 1) consists of a weight A, a non-conducting shaft B and a brass platen C. The total weight in air of the three components is 2 kg. The weight A shall be separated from the platen C on the shaft B by a minimum distance of 75 mm. The shaft shall be hard rubber or a similar poor thermal conductor to avoid loss of heat due to conduction. The diameter of the brass platen which touches the test specimen shall be not less than 50 mm. The thickness of the platen shall not exceed 6.5 mm.

A-2.1.2 Micrometer Caliper - capable of measuring to 0.001 mm.

A-2.1.3 Water-Bath

A-2.2 Preparation of Test Specimen

Break a sufficient number of cakes or sticks of the material into pieces and place in an iron crucible of 60 ml capacity. Place the crucible 130 mm below a 250 W infrared lamp bulb. Heat the stirred compound to a temperature of $75 \pm 5^\circ\text{C}$ (checked with a thermometer) until melted throughout and then pour into a mould (see Fig. 2). The mould shall be a flat, stainless steel metal plate 6.0 mm thick, containing four holes of 10.0 mm in diameter. Preheat the mould to a temperature of $55 \pm 5^\circ\text{C}$ and place on a smooth glass slab (150 mm long, 75 mm wide, 20 mm thick.) preheated to the same temperature. After the mould has been overfilled with the material, place a smooth flat tin-foil-covered glass plate preheated to $55 \pm 5^\circ\text{C}$ on top of the mould. Apply a load of 9 kg to the top of the tin-foiled glass plate for ten minutes. Then place the mould in water at a temperature of 10°C for ten minutes. Remove the upper tin-foiled plate and trim away the excess material. This can be accomplished by using a metal scraper with the top surface of the mould as a guide. Then remove the mould from the glass slab by gently tapping the side of the mould. Remove the specimens from the mould by chilling in water for ten minutes at 10°C and store at room

temperature ($27.0 \pm 2.0^{\circ}\text{C}$) for 24 hours before testing.

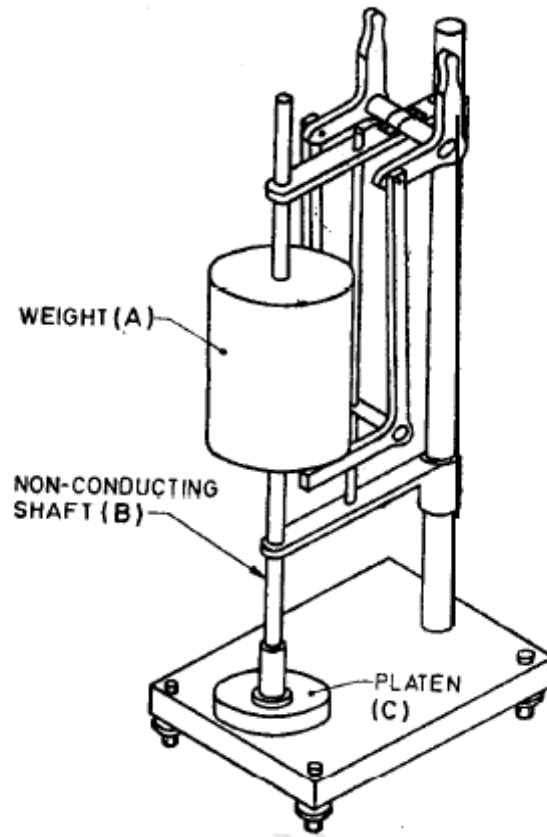


Fig. 1 Flow Testing Machine

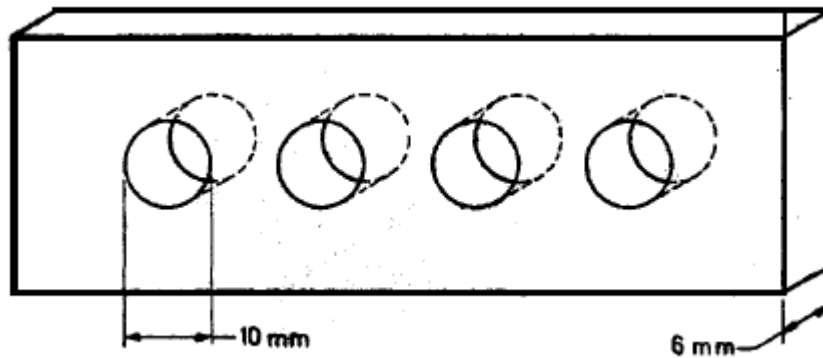


Fig. 2 Mould For Forming Flow Specimens

A-2.3 Procedure

Measure the original length of the specimen using a metric micrometer caliper. Make four measurements around the circumference and one measurement on the centre of the specimen. Record the measurements to the nearest 0.005 mm. Place the specimen and flow testing instrument (see Fig. 1) in a water-bath and hold at the testing temperature for 20 minutes prior to testing. Control the temperature in the bath to within $\pm 0.1^{\circ}\text{C}$ of the required temperature (use a calibrated thermometer for determining the temperature). Place a thin sheet of waterproof cellophane between the instrument and each end of the specimen. The bottom of the specimen shall be 50 mm below the surface of the water in the bath. Apply a constant axial load of 2 kg to the specimen for ten minutes, after which remove the specimen and cool in air to room temperature. Strip off the cellophane and determine the final length as in the case of the original length.

A-2.4 Calculation

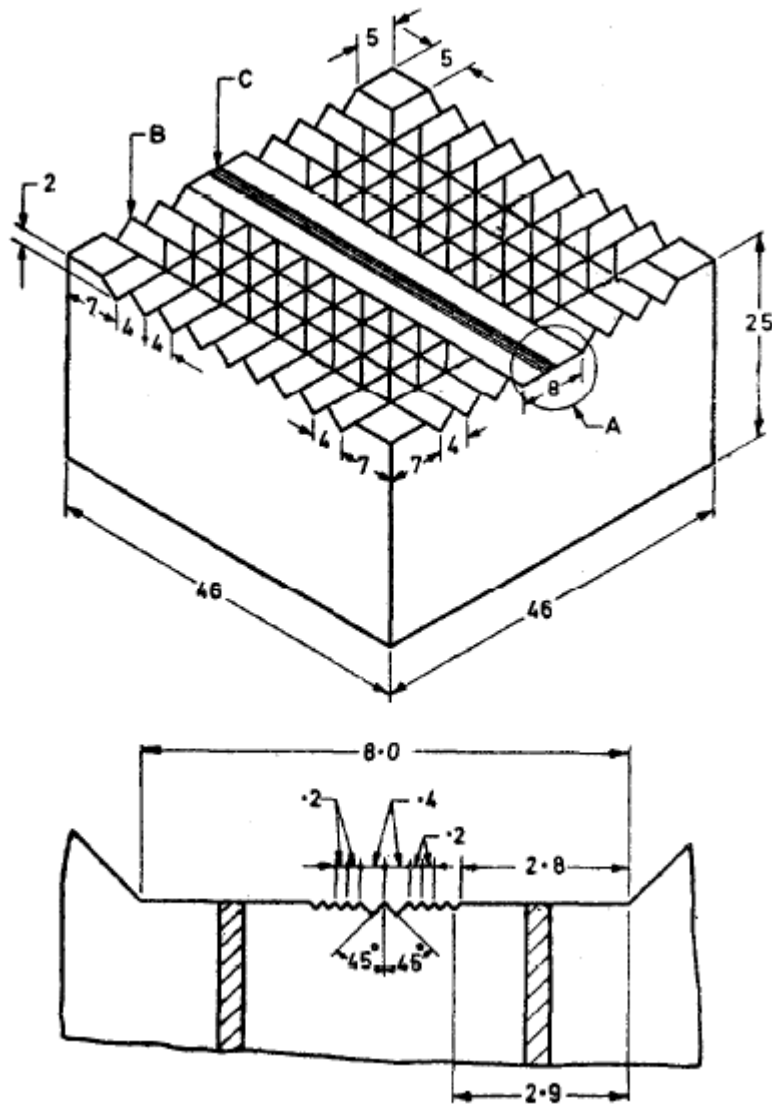
Calculate flow as evidenced by the change in length as percentage of the original length. Report the mean of the values for two specimens to the nearest 0.1 percent.

A-3 IMPRESSION TEST

A-3.1 Apparatus

A-3.1.1 Test Block - See Fig. 3.

A-3.1.2 Brass Weight - weighing $1.000 \pm 5\text{g}$ in air, and having a base 50 mm.



ENLARGED VIEW AT A
All dimensions in millimetres
Fig. 3 ,Block For Impression Test

Note-All grooves, 90° included angle, sides at 45° to surface of the block.

A-3.1.3 Water-Bath

A-3.2 Preparation of Test Specimens

The specimen shall be a disc 40 mm in diameter and between 4 mm and 7 mm in thickness. Warm the cake of compound in water and cut the disc. If the cake is thinner than 4 mm, flame and press together two cakes before cutting the disc.

A-3.3 Procedure

Adjust a water-bath to $45^{\circ}0 \pm 0^{\circ}1^{\circ}\text{C}$. Place in the bath a flat plate (to support the specimens while it is coming to the required temperature), the test block (see Fig. 3) and the brass weight, and allow to come to the temperature of the bath. Keep the water level in the bath 30 ± 5 mm above the top surface of the test block. Then place the specimen, in the water-

bath, on the flat plate. Twenty minutes later, centre the specimen on the test block, place a sheet of waterproof cellophane over the specimen, and place the weight on the specimen. Ten minutes later, remove the weight, take out the test block and specimen from the bath and chill at a temperature of 10°C. Remove the specimen from the test block.

A-3.3.1 Examine the impression on the-specimen to determine if the ridges corresponding to the large cross grooves (B in Fig. 3) of the test block are sharp, and the ridges corresponding to the fine grooves (C in Fig. 3) are complete and visible to the unaided eye for at least 3 mm of their length.

APPENDIX B
(Clause 6)

SAMPLING

B-1. GENERAL REQUIREMENTS OF SAMPLING

B-1.0 In drawing, preparing, storing and handling test samples, the precautions and directions given in B-1.1 to B-1.5 shall be observed.

B-1.1 Samples shall not be taken in an exposed place.

B-1.2 Sampling instruments shall be clean and dry.

B-1.3 Precautions shall be taken to protect the samples, the material being sampled, the sampling instrument and the containers for samples from adventitious contamination.

B-1.4 The sample shall be placed in clean, dry, air-tight glass or other suitable containers.

B-1.5 Each sample container shall be sealed air-tight with a suitable stopper after filling, and marked with full details of sampling, the date of sampling and the year of manufacture of the material.

B-2 SCALE OF SAMPLING

B-2.1 Lot

All the containers in a single consignment of the material drawn from a single batch of manufacture shall constitute a lot. If a consignment is declared or known to consist of different batches of manufacture, the containers belonging to the same batch shall be grouped together and each such group shall constitute a separate lot.

B-2.1.1 Samples shall be tested from each lot for ascertaining conformity of the material to the requirements of the specification.

B-2.2 The number of containers (n) to be selected from the lot shall depend on the size of the lot (N) and shall be as given in Table 2.

TABLE 2 NUMBER OF CONTAINERS TO BE SELECTED FOR SAMPLING
(Clause B-2.2 and B-3.1.1)

Lot Size N	No. of Containers to be Selected n
(1)	(2)
3 to 50	3
51 to 200	4
201 to 400	5
401 to 650	6
651 to 1000	7
1001 and above	8

B-2.3 These Containers shall be selected from the lot at random and in order to ensure randomness of selection, procedures given in IS :4905.

B-3. TEST SAMPLES AND REFEREE SAMPLE

B-3.1 Preparation of Test Samples

B-3.1.1 Draw with an appropriate sampling instrument a small portion of the material from different parts of each container selected (*see* Table 2). The total quantity of the material drawn from each container shall be sufficient to conduct the tests for all the characteristics given under 3 and shall be not less than 250 g.

B-3.1.2 Thoroughly mix all portions of the material drawn from the same container. Out of these portions, equal quantities shall be taken from each selected container and shall be well mixed up together so as to form a composite sample weighing not less than 0.5 kg. This composite sample shall be divided into three equal parts, one for the purchaser, another for the supplier and the third for the referee.

B-3.2 Referee Sample

The referee sample shall consist of a composite sample marked for this purpose and shall bear the seals of the purchaser and the supplier. It shall be kept at a place agreed to between the purchaser and the supplier and shall be used in case of dispute between the two.

B-4. NUMBER OF TESTS

B-4.1 Tests for all the characteristics given in clause 4 shall be conducted on the composite sample.

B-5. CRITERIA FOR CONFORMITY

B-5.1 A lot shall be declared as conforming to this specification if the composite sample satisfies the requirements for each of the characteristics given in clause 4. If the requirements for any of the characteristics are not met, the lot shall be declared to have not satisfied the requirements of the specification.